

ACADEMIC DETAILS

Education / Degree	Institute	Year	CPI/%
Ph.D., Materials Science & Engineering	IIT Kanpur	2021	8.5/10
M.Tech, Materials Science & Engineering	IIT Jodhpur	2021	9.03/10
B.E, Mechanical Engineering	Institute of Technology Korba, Chhattisgarh	2017	76.67%
Class XII (CGBSE)	Shiva Public School, Bhilai	2013	79%

TECHNICAL SKILLS

Languages	Libraries	Softwares
Python, bash-scripting	Numpy, Scikit, Matplotlib, GNU-plot, Xmgrace, Phonopy API, pymatgen, OVITO Python API, SQLite3, Paramiko (SSH), PyYAML	VASP, Quantum ESPRESSO, QuantumATK, LAMMPS, PFDD, DAMASK, CALPHAD, VESTA, Ovito, Ollama (local LLM inference), atomicrex

ACHIEVEMENT

- Awarded Prime Minister’s Research Fellowship (PMRF):** Honored to be recognized as a top research talent in India through this prestigious fellowship program. The PMRF supports my research potential and fuels my passion for innovation. Aug’22-July’25
- GATE Qualified:** Demonstrated strong fundamentals in engineering through qualifying for the Graduate Aptitude Test in Engineering (GATE). This national exam assesses a comprehensive understanding of engineering principles. Feb. 2019, ME

WORK EXPERIENCE

- Scale AI, Human Frontier Collective (HFC)** May’25-Oct.25
 - Worked for Scale AI’s prestigious Human Frontier Collective, a part-time internship for leading researchers (PhDs, postdocs).
 - Collaborated on cutting-edge generative AI research initiatives, with exclusive access to advanced proprietary models.
 - Contributed to rigorous testing, critical analysis, and strategic direction of transformative next-generation AI systems.

PROJECTS UNDERTAKEN, IIT KANPUR

- Determination of Dihedral Angles at Grain Boundary Triple Junctions Using Image Processing Technique** Sept’21-Nov’21
 - This project was completed under the supervision of Prof. Sandeep Sangal. In this study, I used an image processing technique to determine the location of grain boundaries, triple point junctions, and the dihedral angle at triple point junctions in a polycrystalline microstructure.
 - While doing this project, I explored several libraries like Scikit, Numpy, Matplotlib, ndimage, etc.
- Determining the relationship between d-band width and Stacking fault Energy of all Transition Metals and Alloys** Ongoing
 - This project is running under the supervision of Prof. Somnath Bhowmick. In this, we are trying to corelate the width of the d-band and stacking fault energy (SFE). The idea is to get the qualitative SFE value by directly looking into the partial density of state (PDOS) of any transition metals and alloys.
- A Computational Study on Borophene to Check its Potential as an Anode Material for Na-ion Battery** Sept’21-Nov’21
 - Under the supervision of Prof. Somnath Bhowmick, I conducted a detailed computational study based on density functional theory to investigate the structural, energy storage capacity, and electronic properties of borophene and determine whether it can be a promising anode material for Na-ion battery applications.
- Computational Investigation of Titanium: Bridging Length Scales from Electronic to Continuum** Ongoing
 - This project is running under the guidance of Prof. Somnath Bhowmick and Prof. N P Gurao.
 - Conducted DFT calculations with VASP to determine key quantum mechanical properties of Titanium, forming the basis for subsequent MD simulations. Performed MD simulations using LAMMPS to study dislocation mobility and grain boundaries under varying conditions, integrating results from DFT calculations. Utilized DD and CPFFT simulations to connect atomic-level insights from MD to macroscopic behaviors, providing comprehensive understanding of Titanium’s mechanical response
- An Autonomous Multi-Scale ICME Agent Framework: Seamless Integration of Ab Initio DFT, Molecular Dynamics, and Crystal Plasticity Finite Element Simulations via Automated Scale-Bridging** Ongoing
 - This project is carried out under the supervision of Prof. Somnath Bhowmick and Prof. N P Gurao. The work involves the design and implementation of a fully autonomous multi-scale simulation framework that integrates VASP (DFT, Ångström scale), LAMMPS (molecular dynamics, nanometre scale), and DAMASK (crystal plasticity finite element, micron-to-millimetre scale) within a single orchestrated pipeline accessible via a natural-language interface.
 - Designed and implemented three automated scale-bridge modules (VASP→LAMMPS, VASP→DAMASK, and LAMMPS→DAMASK) that extract, unit-convert, and inject 18+ material parameters, including elastic constants, stacking-fault energies, dislocation drag coefficients, and Burgers vector magnitudes, between simulation codes without any manual intervention, along with a full provenance-tracking system for every transferred parameter.
 - Built a library of 85 pre-built simulation workflows across the three codes (24 VASP, 30 LAMMPS, 31 DAMASK), a self-healing error-recovery system recognising 13 VASP and 8 LAMMPS error signatures, an automated convergence-testing module, a Jaccard-similarity-based structure memory cache, and a SQLite results database with cache-skip logic, collectively reducing end-to-end simulation setup time by more than 60× compared to manual expert workflows.
 - Validated the complete pipeline on Al (FCC) and Ti (HCP): DFT elastic constants agreed with Materials Project reference values to within 1%; DAMASK-predicted yield stress for Ti HCP polycrystal agreed with experiment to within 8% using exclusively computationally derived constitutive parameters.

JOURNAL PUBLICATIONS

1. **Md Faiz Akhtar**, Nilesh P Gurao, Somnath Bhowmick "Predicting Primary Slip Systems in Hcp Metals: A Combined Geometrical and Electronic Approach" Vol. 400, 1 Dec. 2025, 139019 DOI: 10.1016/j.matlet.2025.139019
2. **Faiz Akhtar, Md**, Nilesh P. Gurao, and Somnath Bhowmick. "A novel electronic structure-based prediction of interstitial strengthening in α -titanium." Journal of Applied Physics 138.17 (2025)link.
3. A Linda, **Md Faiz Akhtar**, Shaswat Pathak, Somnath Bhowmick "Phys. Rev. B 109, 214102 "Accelerating Generalized Stacking Fault Energy Prediction by Combining Friedel Model, Ab Initio Calculation and Machine Learning" DOI:10.1103/PhysRevB.109.214102
4. Umashankar Rajput, **Md Faiz Akhtar**, Imran khan Mulani, Giriraj Vyas, Shailendra, K. Chaubey, Somnath Bhowmick, Prashant Kumar, and Aparna Deshpande "Liquid Phase Exfoliated Borophene on Au(111)" DOI:10.1021/acs.jpcc.4c00497
5. Umashankar Rajput, **Md Faiz Akhtar**, Imran khan Mulani, Giriraj Vyas, Shailendra, K. Chaubey, Somnath Bhowmick, Prashant Kumar, and Aparna Deshpande "Evolution of atomistic boron clusters on borophene monolayers on Au(111)" DOI:10.1021/acs.jpcc.5c00397
6. A Linda, **Md Faiz Akhtar**, Somnath Bhowmick "Deformation in Metals: Insights from ab-initio Calculations " DOI:10.1007/978-981-99-6863-3_10
7. **Md Faiz Akhtar**, Jiajie Zhu, Appala Naidu Gandhi "Functionalized Ti_2C Based MXenes For Li-ion Battery Applications" *Manuscript prepared*
8. **Md Faiz Akhtar**, Nilesh P Gurao, Somnath Bhowmick "Computational Investigation of Titanium: Bridging Length Scales from Electronic to Continuum" *Under Preparation*
9. **Md Faiz Akhtar**, Nilesh P Gurao, Somnath Bhowmick "An Autonomous Multi-Scale ICME Agent Framework: Seamless Integration of Ab Initio DFT, Molecular Dynamics, and Crystal Plasticity Finite Element Simulations via Automated Scale-Bridging" *Computational Materials Science (Under Preparation)*

CONFERENCES & SYMPOSIUM ATTENDED

- **International Symposium on Transformational Technologies in Materials Manufacturing**, "Effect of interstitial atoms on the deformation behavior of titanium: an Ab-initio study", University of Agricultural Sciences, Bengaluru, on 20-22 November 2024
- **30th International Conference on Computational & Experimental Engineering and Sciences**, "Computational Investigation of Titanium: Bridging Length Scales from Electronic to Continuum," Singapore on August 3-6, 2024
- **HPC Symposium-2024**, "Computational Investigation of Titanium: Bridging Length Scales from Electronic to Continuum," IIT Kanpur on 20th April 2024
- **PMRF symposium-2024**, "Accelerating the prediction of stacking fault energy by combining ab initio calculations and machine learning," IIT Indore on March 3rd and 4th, 2024
- **IIM-ATM 2023**, "Predicting primary slip systems in hcp materials: A combined geometrical and electronic approach, KIIT University, Bhubaneswar, on 22 - 24 November 2023
- **METCENT-2023**, "A critical role of oxygen on the deformation behaviour of Ti and Ti-Al alloys" IIT BHU on 26-28 October 2023

School Attended

- **LOBSTER School**, attended LOBSTER School on chemical bonding analysis from 11 to 13 June 2025, organized by BAM Akademie, Berlin, Germany.
- **GIAN**, attended Modelling Defects in Crystalline Solids in Jan 2025, Instructor Prof. David Rodney, organised by IIT Madras.

Positions of responsibilities, and EXTRA-CURRICULAR ACTIVITIES

- **Overall Ph.D. Placement Coordinator**, Students' Placement Office (SPO), IIT Kanpur, 2025-26. Spearheading placement and internship activities for the entire Ph.D. cohort across all departments, coordinating with recruiters from industry and R&D organizations, managing student profiles, organizing pre-placement talks, and serving as a key liaison between Ph.D. scholars and prospective employers to maximize placement outcomes.
- **President** student body, IIM Kanpur chapter, IIT Kanpur. Lead the IIM Kanpur Chapter, organized events such as Metals and Materials Quiz, IIM Foundation Day, Annual General Meeting (AGM), workshops, conferences, and lectures to promote materials and metallurgical engineering awareness.
- **Overall-Coordinator** Materials Science society, MatSoc, MSE Departement IIT Kanpur. As the overall coordinator of MatSoc at IIT Kanpur, our team lead a dynamic student community dedicated to advancing learning, research, and innovation in materials science. We organize events, workshops, and guest lectures, sports and fun activities providing students with access to cutting-edge research facilities and opportunities for professional growth.
- **Senator**, student senate IIT Kanpur, as a Senator I am representing the Ph.D. students at IIT Kanpur, I advocate for student interests, contribute to policy-making, and facilitate communication between students and administration to enhance the academic and campus experience.
- **Student Seceratory**, National Symposium of Research Scholars on Metallurgy and Metals, NSRS-2024, held at IIT Kanpur. Organized and facilitated NSRS-2024, a national event fostering collaboration and knowledge sharing among research scholars in metallurgy and materials science, featuring workshops, plenary lectures, and presentations by leading experts.
- **Institute Badminton team member**, represent IIT Kanpur in UDGHOSH22, 23, and 24